

TASKS'

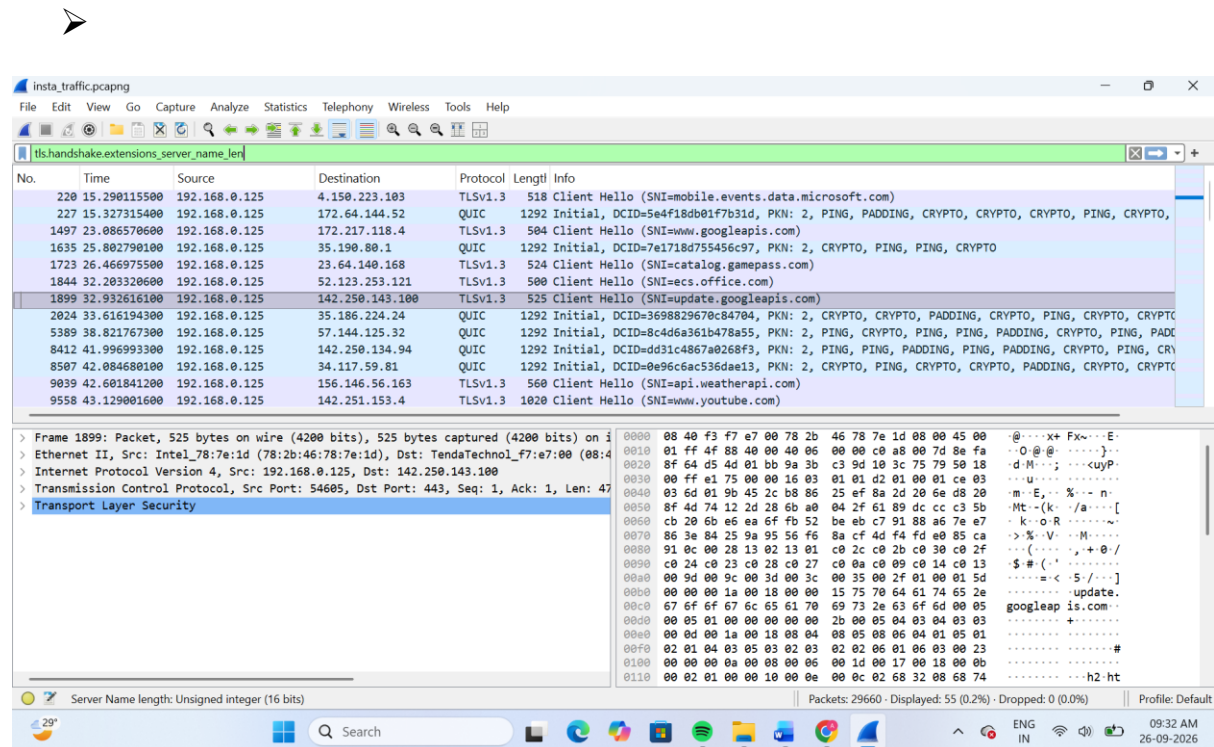
1. Install Wireshark on your computer and capture 2 minutes of network traffic while you browse Instagram or YouTube. Save the capture file as insta_traffic.pcap.



Wireshark interface showing a network capture file named 'insta_traffic.pcapng'. The main display area shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The packet list shows several TCP and PSH/ACK packets between 192.168.0.125 and 34.104.35.123. The packet details pane on the right shows the selected packet (No. 1) with its structure: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Transport Layer Security. The status bar at the bottom indicates 'Ready to load or capture' and 'Packets: 29660 - Dropped: 0 (0.0%)'.

No.	Time	Source	Destination	Protocol	Length	Info
29649	69.621443700	142.251.153.4	192.168.0.125	TCP	54	443 → 60807 [ACK] Seq=1352794 Ack=373848 Win=348928 Len=0
29650	69.621443900	142.251.153.4	192.168.0.125	TCP	54	443 → 60807 [ACK] Seq=1352794 Ack=375932 Win=353024 Len=0
29651	69.622831300	34.104.35.123	192.168.0.125	TCP	1506	80 → 54606 [ACK] Seq=16001965 Ack=332 Win=268800 Len=1452
29652	69.622885200	192.168.0.125	34.104.35.123	TCP	54	54606 → 80 [ACK] Seq=332 Ack=16003417 Win=65280 Len=0
29653	69.625032500	34.104.35.123	192.168.0.125	TCP	1506	80 → 54606 [PSH, ACK] Seq=16003417 Ack=332 Win=268800 Len=1452
29654	69.627562100	34.104.35.123	192.168.0.125	TCP	1506	80 → 54606 [ACK] Seq=16004869 Ack=332 Win=268800 Len=1452
29655	69.627617900	192.168.0.125	34.104.35.123	TCP	54	54606 → 80 [ACK] Seq=332 Ack=16006321 Win=65280 Len=0
29656	69.630110200	34.104.35.123	192.168.0.125	TCP	1506	80 → 54606 [PSH, ACK] Seq=16006321 Ack=332 Win=268800 Len=1452
29657	69.632458200	34.104.35.123	192.168.0.125	TCP	1506	80 → 54606 [ACK] Seq=16007773 Ack=332 Win=268800 Len=1452
29658	69.632531600	192.168.0.125	34.104.35.123	TCP	54	54606 → 80 [ACK] Seq=332 Ack=16009225 Win=65280 Len=0
29659	69.634547100	34.104.35.123	192.168.0.125	TCP	1506	80 → 54606 [PSH, ACK] Seq=16009225 Ack=332 Win=268800 Len=1452
29660	69.641790200	34.104.35.123	192.168.0.125	TCP	1506	80 → 54606 [ACK] Seq=16010677 Ack=332 Win=268800 Len=1452

2. Use Wireshark to filter and list all HTTP or HTTPS requests made by your browser during the capture. Write down the domains you see most frequently.



Wireshark was used to filter HTTP/HTTPS traffic generated by the browser. The captured traffic mainly contained HTTPS/TLS connections. The domains observed during the capture included:

1. www.youtube.com
2. update.googleapis.com
3. www.googleapis.com
4. mobile.events.data.microsoft.com
5. api.weatherapi.com

Result:

The browser generated multiple HTTPS requests during browsing, and the above domains were observed in the captured traffic.

**3.Open your insta_traffic.pcap in Wireshark and find a TCP packet.
Document the source IP, destination IP, and TCP port numbers for that packet.**



Field	Value
Source IP	34.104.35.123
Destination IP	192.168.0.125
Source Port	80
Destination Port	54606
Protocol	TCP

A TCP packet was identified in Wireshark. The source IP address is 34.104.35.123 and the destination IP address is 192.168.0.125. The source port is 80 and the destination port is 54606.

4.Run an Nmap scan on your own machine (localhost) to detect open ports.
Save the command you used and list the open ports
found.

Hint: Use 'nmap 127.0.0.1' in your
terminal.



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>nmap --version
Nmap version 7.991 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.21 nmap-libssh2-1.11.1_NMAP1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.88 nmap-libdnet
-1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select

C:\Users\HP>nmap 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-26 09:45 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
1521/tcp   open  oracle
2179/tcp   open  vmrpd
8080/tcp   open  http-proxy
16992/tcp  open  amt-soap-http

Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds
C:\Users\HP>
```

nmap 127.0.0.1

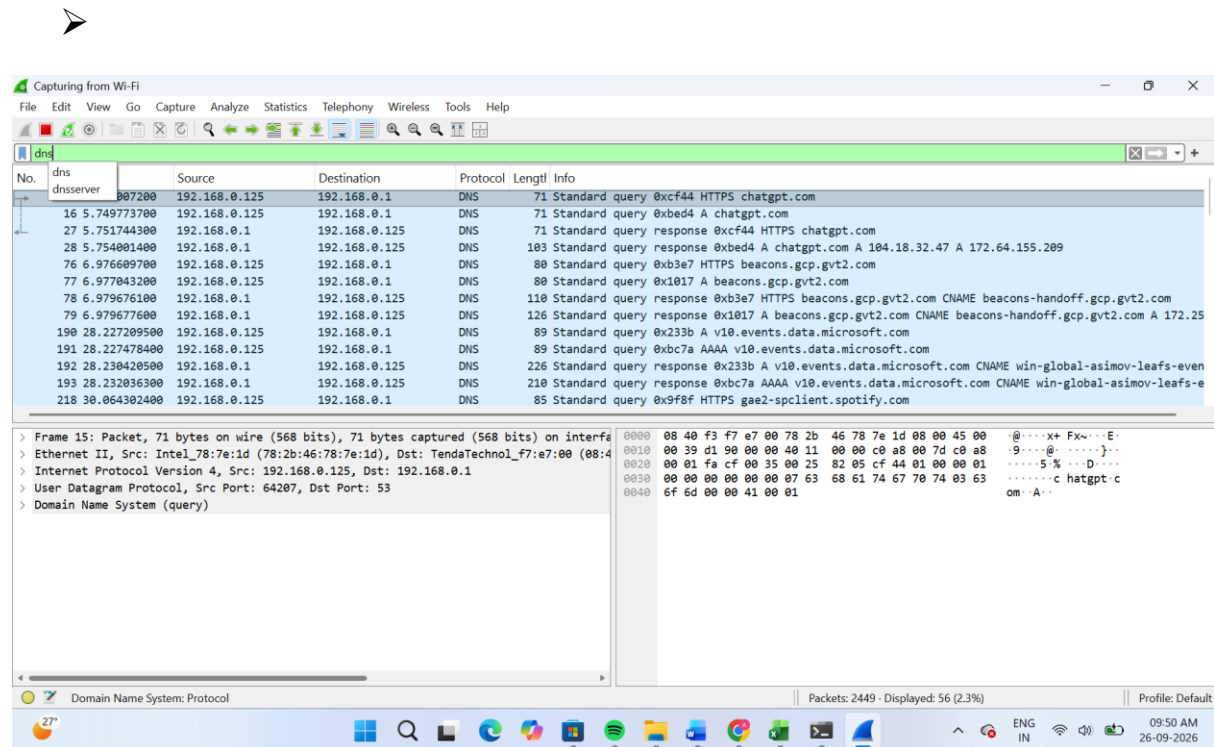
Open Ports Found

- 135/tcp – msrpc
- 445/tcp – microsoft-ds
- 1521/tcp – oracle
- 2179/tcp – vmrpd
- 8080/tcp – http-proxy
- 16992/tcp – amt-soap-http

Result

Nmap scan was successfully performed on localhost (127.0.0.1), and six open TCP ports were detected.

5. Capture network traffic while using a food delivery app (like Zomato or Swiggy) for 1 minute. In Wireshark, identify at least one DNS request and document the domain name being resolved. **Hint:** Use the 'dns' filter in Wireshark to find DNS packets.



The network traffic was captured for one minute, and the DNS request for chatgpt.com was identified using Wireshark.